

where the inequality follows from the general fact that $(K + C) \setminus (L + C) \subset (K \setminus L) + C$. Moreover, if A, B, C are convex, compact sets then the estimate is non-trivial, i.e., using Theorem 3.9 we get that the right hand side of the above quantity is non-negative. It is interesting to note that the Δ operation is not commutative, i.e. $\Delta_C \Delta_B(A) \neq \Delta_A \Delta_C(A)$; this can be seen, for example, in $\mathbb{R}^2$ by taking A to be a square, B to be a segment, and C to be a Euclidean ball.

It is natural to ask if the higher-order analog of this observation remains true.

Question 3.10. *Let $m \in \mathbb{N}$ and $B_1, \dots, B_m \subset \mathbb{R}^n$ be compact sets containing the origin. Then, for any compact $B_0 \subset \mathbb{R}^n$, is it true that*

$$|\Delta_{B_1} \dots \Delta_{B_m}(B_0)| \geq \sum_{s \subset [m]} (-1)^{m-|s|} \left| B_0 + \sum_{i \in s} B_i \right|?$$

The inequality (23) gives a positive answer to the above question in the case $m = 2$. We also observe that if $B_0, B_1, \dots, B_m$ are convex, then the right hand side is non-negative thanks to Theorem 3.6. We note that it was observed in [20], by considering $A = \{0, 1\}$ and $B = C = [0, 1]$ in $\mathbb{R}^1$, that the volume of Minkowski sums cannot be supermodular (even in dimension 1) if the convexity assumption on the set A is removed. Nonetheless [20] observed that if $A, B, C \subset \mathbb{R}$ are compact, then

$$|A + B + C| + |\text{conv}(A)| \geq |A + B| + |A + C|;$$

it is unknown if this extends to higher dimension. In particular, we do not know if the following conjecture is true for $n \geq 2$.

Conjecture 3.11. *For any convex body A and any compact sets B and C in $\mathbb{R}^n$,*

$$|A + B + C| + |A| \geq |A + B| + |A + C|.$$

We can confirm Conjecture 3.11 under the assumption that B is a zonoid.

Theorem 3.12. *Assume A is a convex compact set, B is a zonoid and C is any compact set in $\mathbb{R}^n$. Then*

$$|A + B + C| + |A| \geq |A + B| + |A + C|.$$

Proof. By approximation, we may assume that B is a zonotope. Using the definition of mixed volumes (4) and (7) we get that for any convex compact set M in $\mathbb{R}^n$

$$|M + [0, tu]| - |M| = t|P_{u^\perp} M|_{n-1}, \text{ for all } t > 0, u \in S^{n-1}.$$

The above formula can be also proved using a geometric approach and thus studied in the case of not necessary convex M . Indeed, consider a compact set M in $\mathbb{R}^n$, $t > 0$ and $u \in S^{n-1}$, let $\partial_u M$ be the set of all $x \in \partial M$ such that $x \cdot u \geq y \cdot u$, for all $y \in M$ for which $P_{u^\perp} y = P_{u^\perp} x$. Note that

$$(\partial_u M + (0, tu]) \cap M = \emptyset, \text{ but } M \cup (\partial_u M + (0, tu]) \subseteq M + [0, tu].$$

Thus

$$|M + [0, tu]| \geq |M| + t|P_{u^\perp} M|_{n-1}.$$